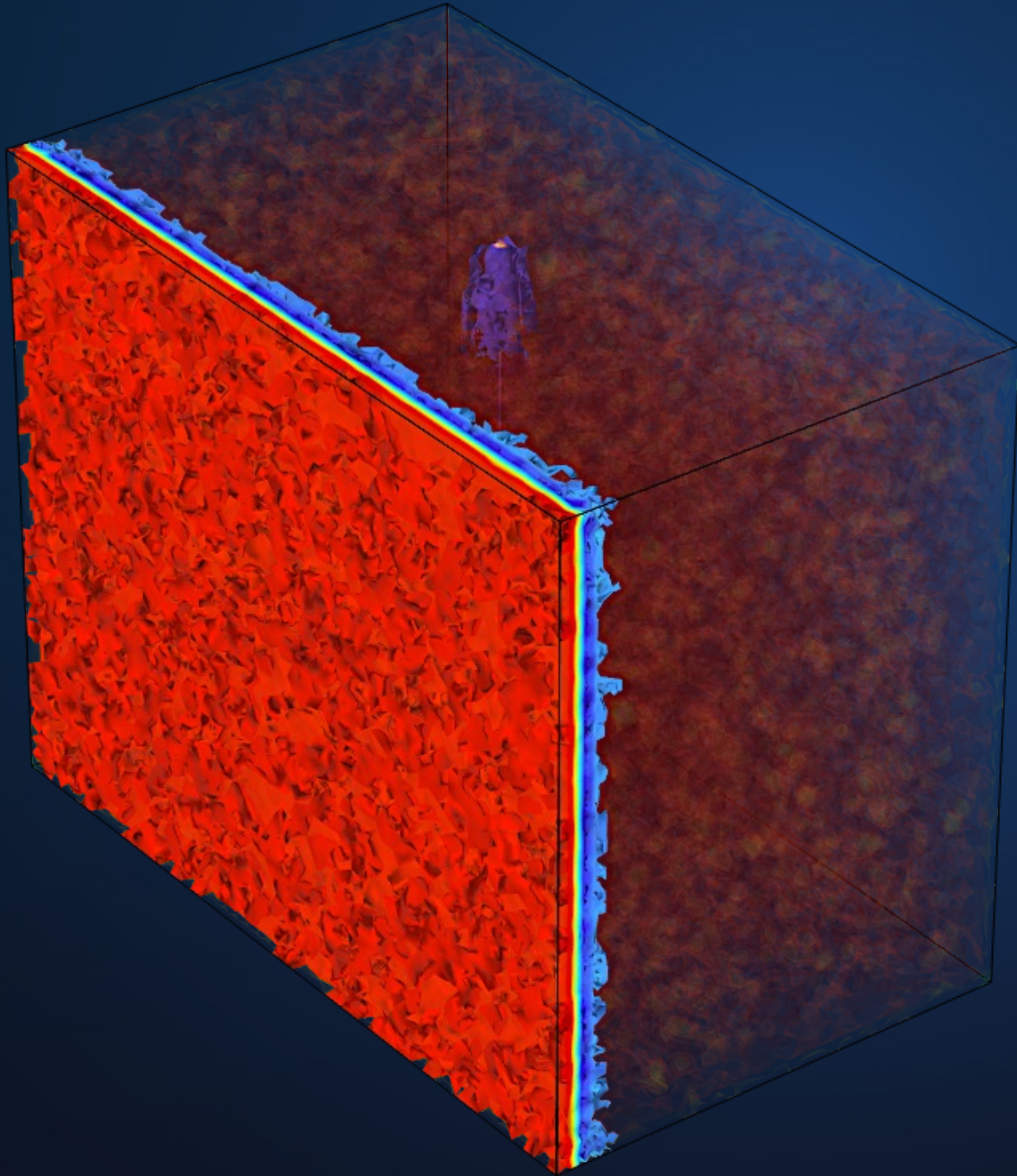
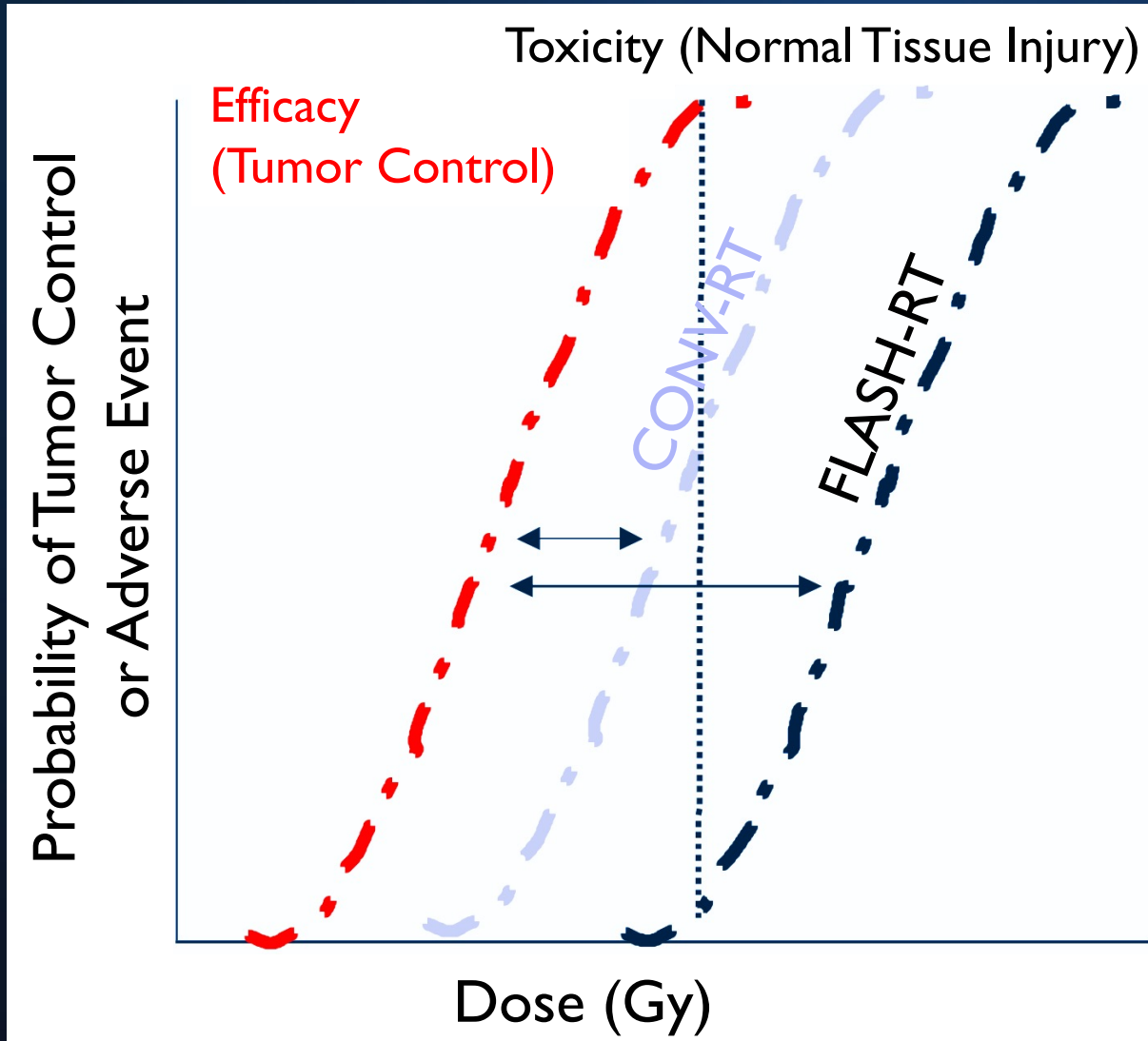




Curing Cancer in a FLASH: A Finite Element-Based Diagnostic Ultrasound Approach to Ultra-High Dose Rate (FLASH) Measurements and Absolute Dosimetry

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Plano East Senior High School

Background



Cengel et al., 2024, p. 219.

Conventional Radiotherapy (CONV-RT):

- 0.01 – 0.40 Gy/s dose rates
- Minute-long pulses
- Relatively High fractions
- Irradiation damage to healthy tissue around a tumor is present

FLASH Radiotherapy (FLASH-RT):

- Ultra-high dose rate (≥ 40 Gy/s)
- Under 200 ms
- Relatively few fractions
- Minimizes irradiation damage to healthy tissue

Shortcomings of FLASH-RT

Problem: Clinical Adoption

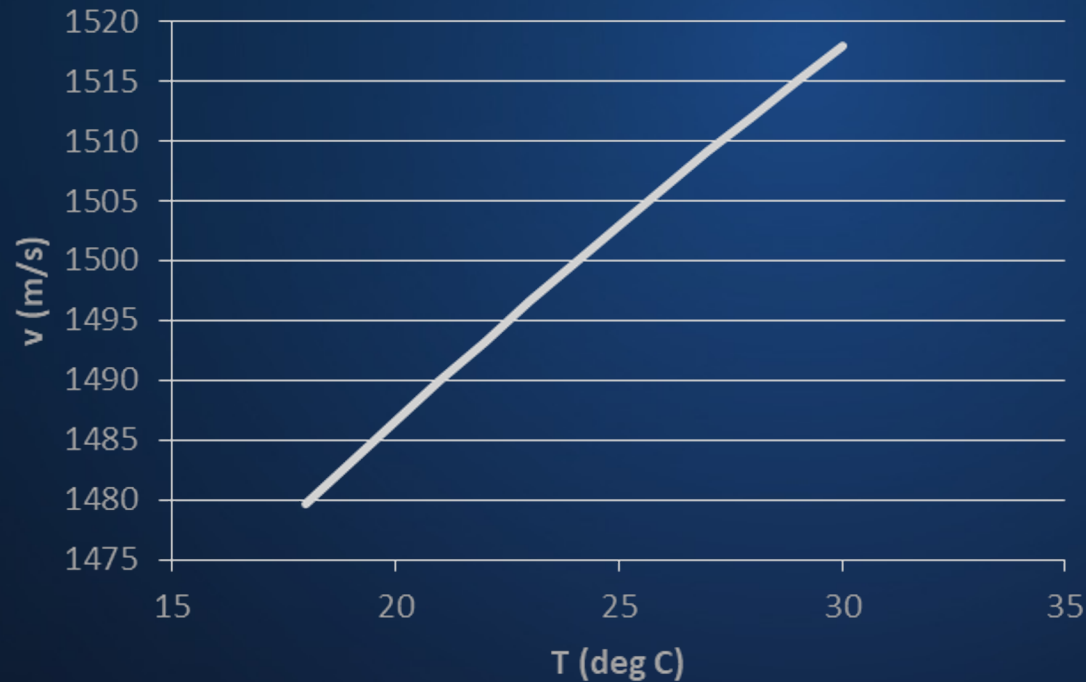
- Need to precisely verify dose rate within a volume
- Current dosimetry hardware and measurement standards are lacking

Solution: Novel Detection Principle

- Detect absolute dose and dose-rate
- Hardware: use of Ultrasound
- Measurement Standard: Phase Modulation

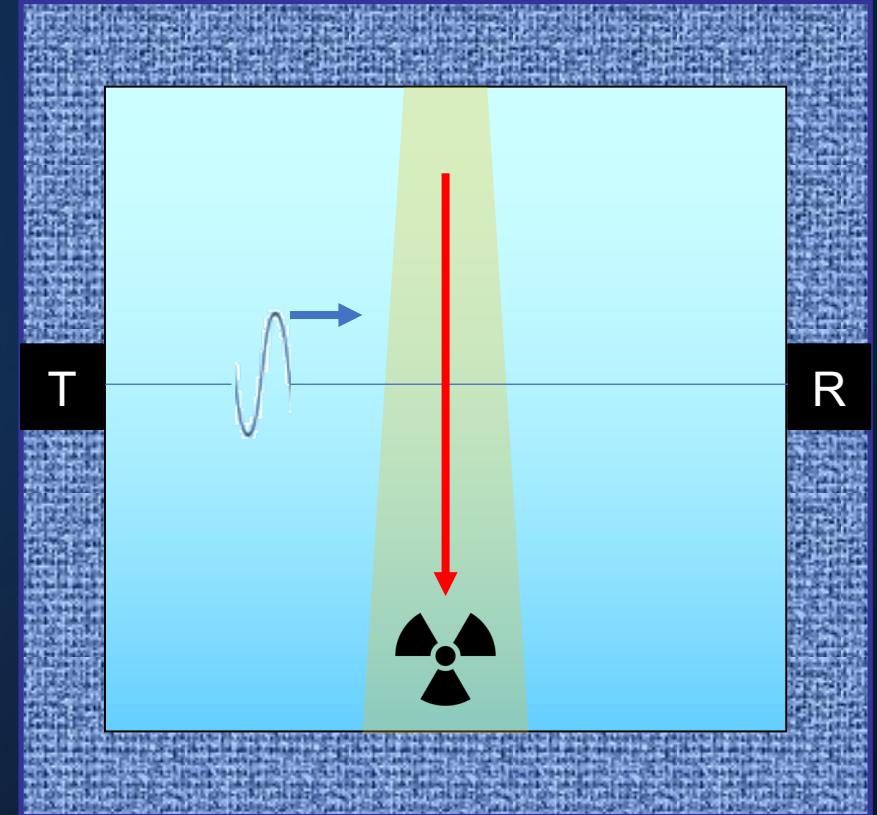
Existing Approach

Speed of Sound in Water vs. Irradiated Temperature



$$v(T) \cong 1394 + 4.5T$$

$$\Delta T \cong 0.2391 \text{ mK/Gy}$$

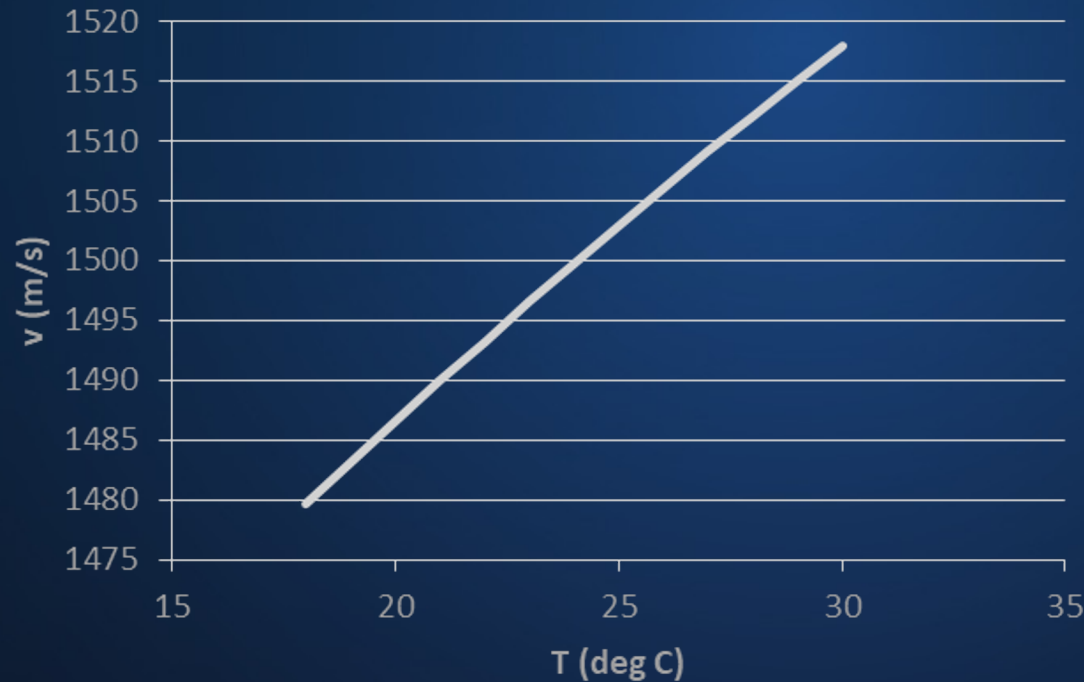


$$TOF \cong 7 \mu\text{s per cm } (\sim 2 \text{ cm})$$

$$\Delta TOF \cong 10 \text{ ps/Gy}$$

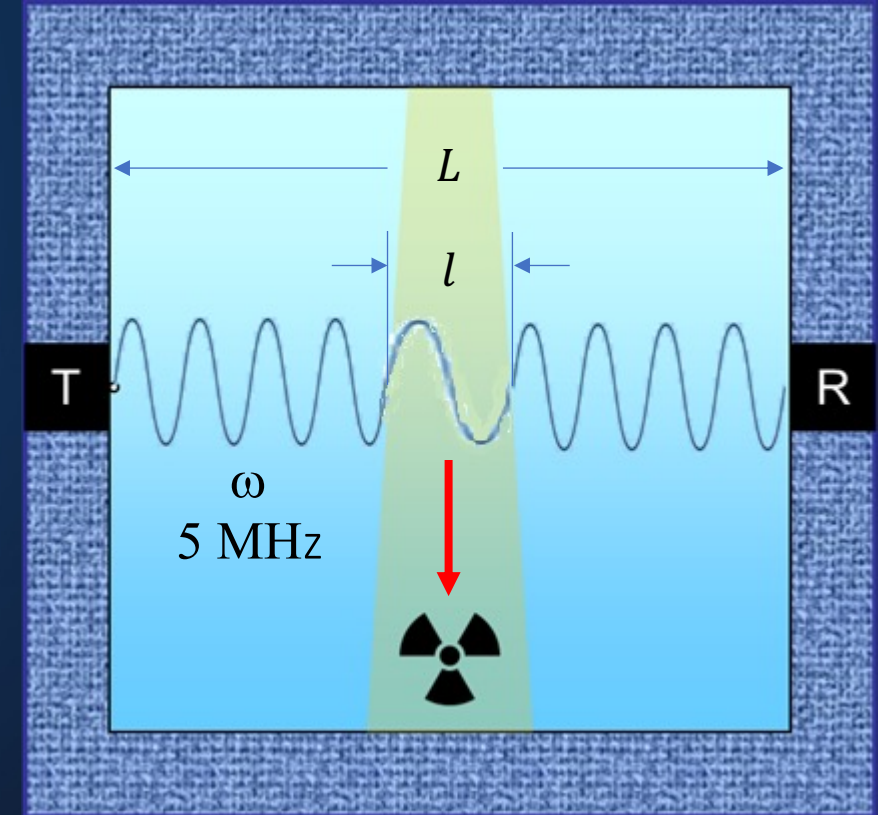
Proposed Approach

Speed of Sound in Water vs. Irradiated Temperature



$$v(T) \cong 1394 + 4.5T$$

$$\Delta T \cong 0.2391 \text{ mK/Gy}$$



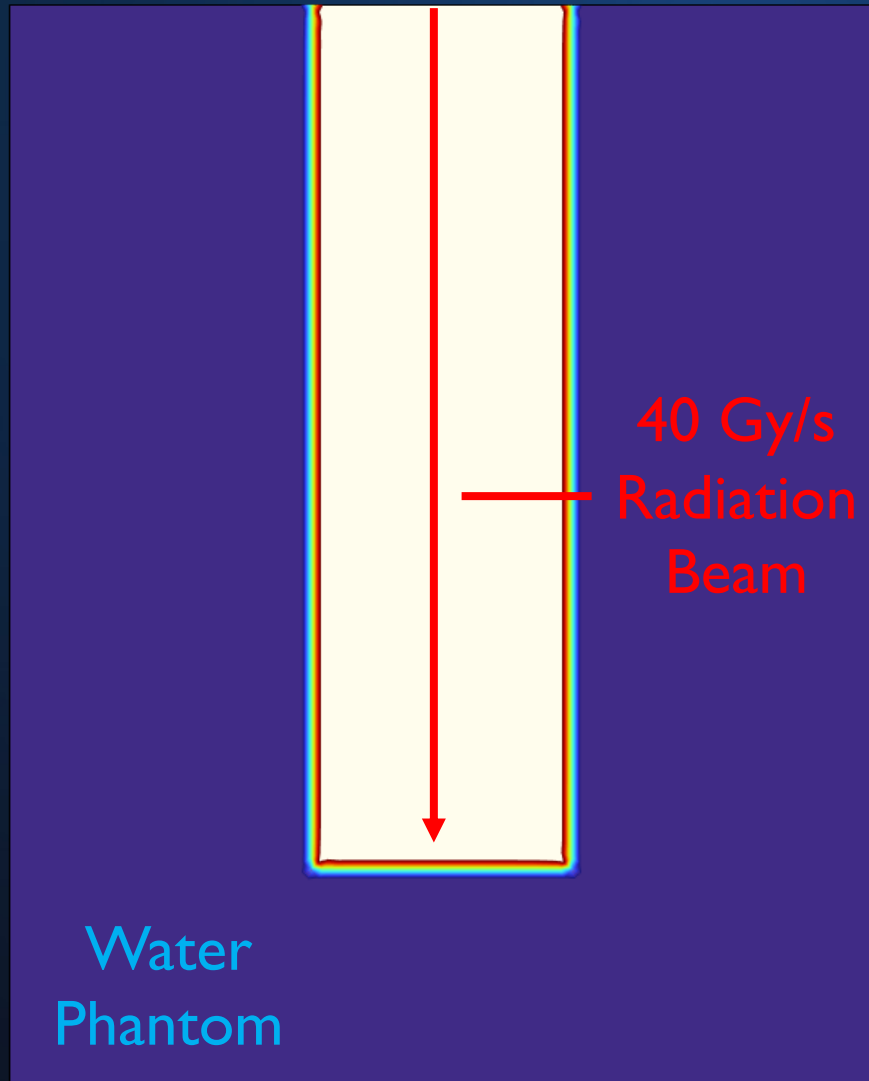
$$\Delta\varphi = \Delta(\omega \cdot TOF)$$

$$\cong \omega \cdot \Delta TOF$$

$$\cong 18 \text{ mdeg/Gy}$$

Thermal Model Validation

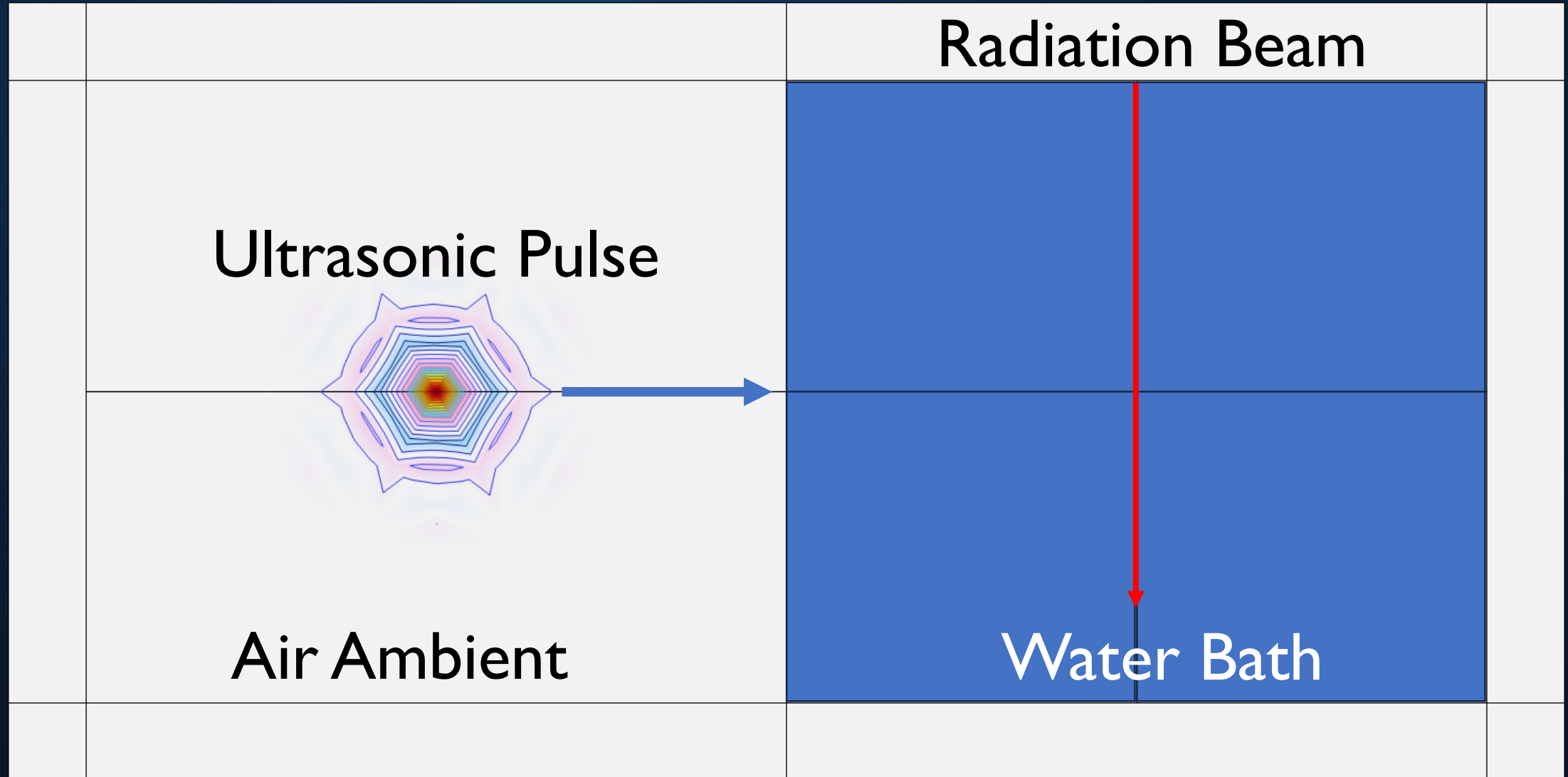
Temperature Contour Plot



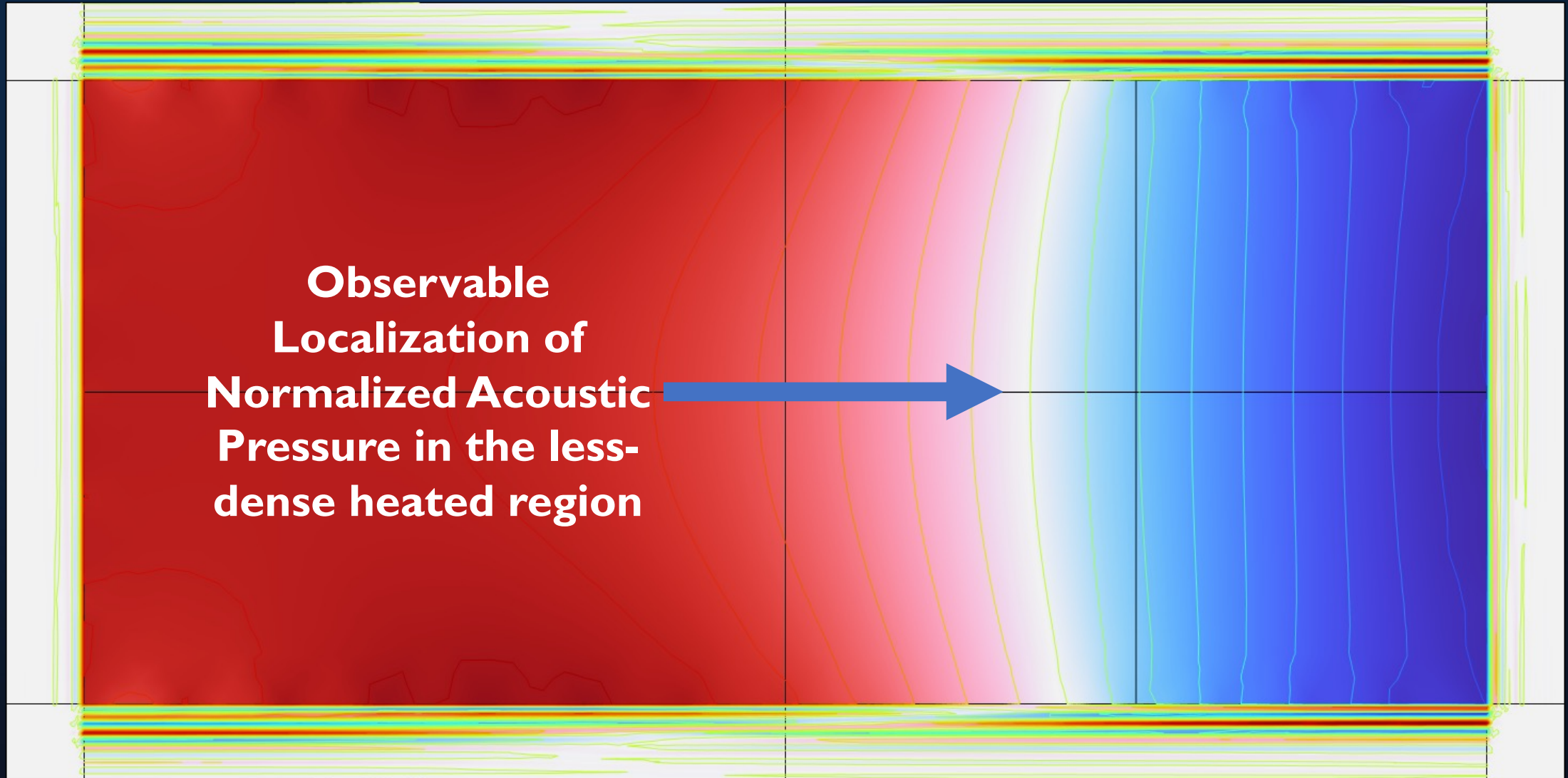
Hotter Temperatures:
Lighter Colors
(Purple = Water Phantom)

Sub-Kelvin heating:
 $\cong 10$ mK temperature difference

Acoustics Model Validation – 1 μ s

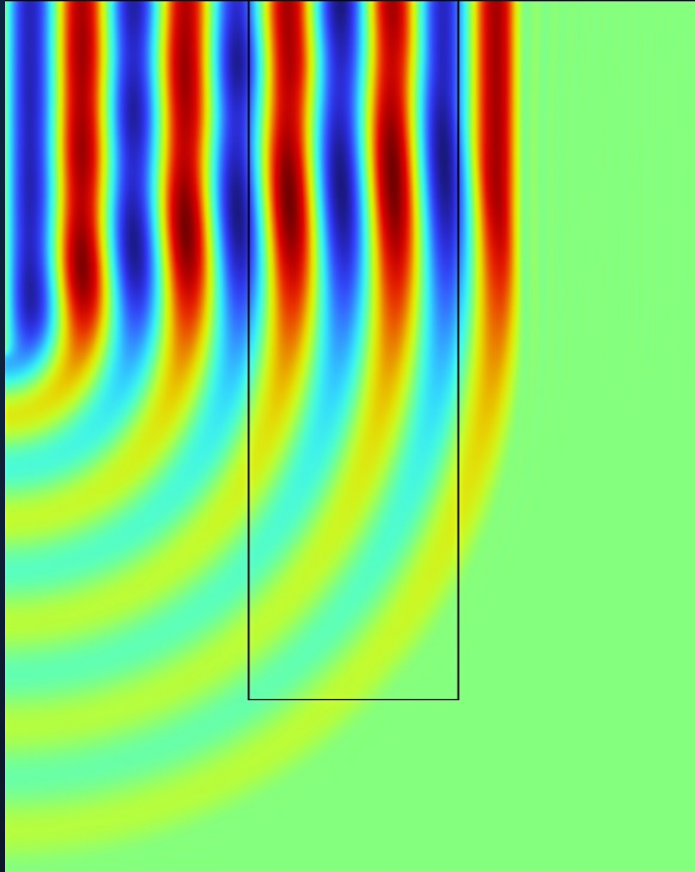


Acoustics Model Validation – 60 μs

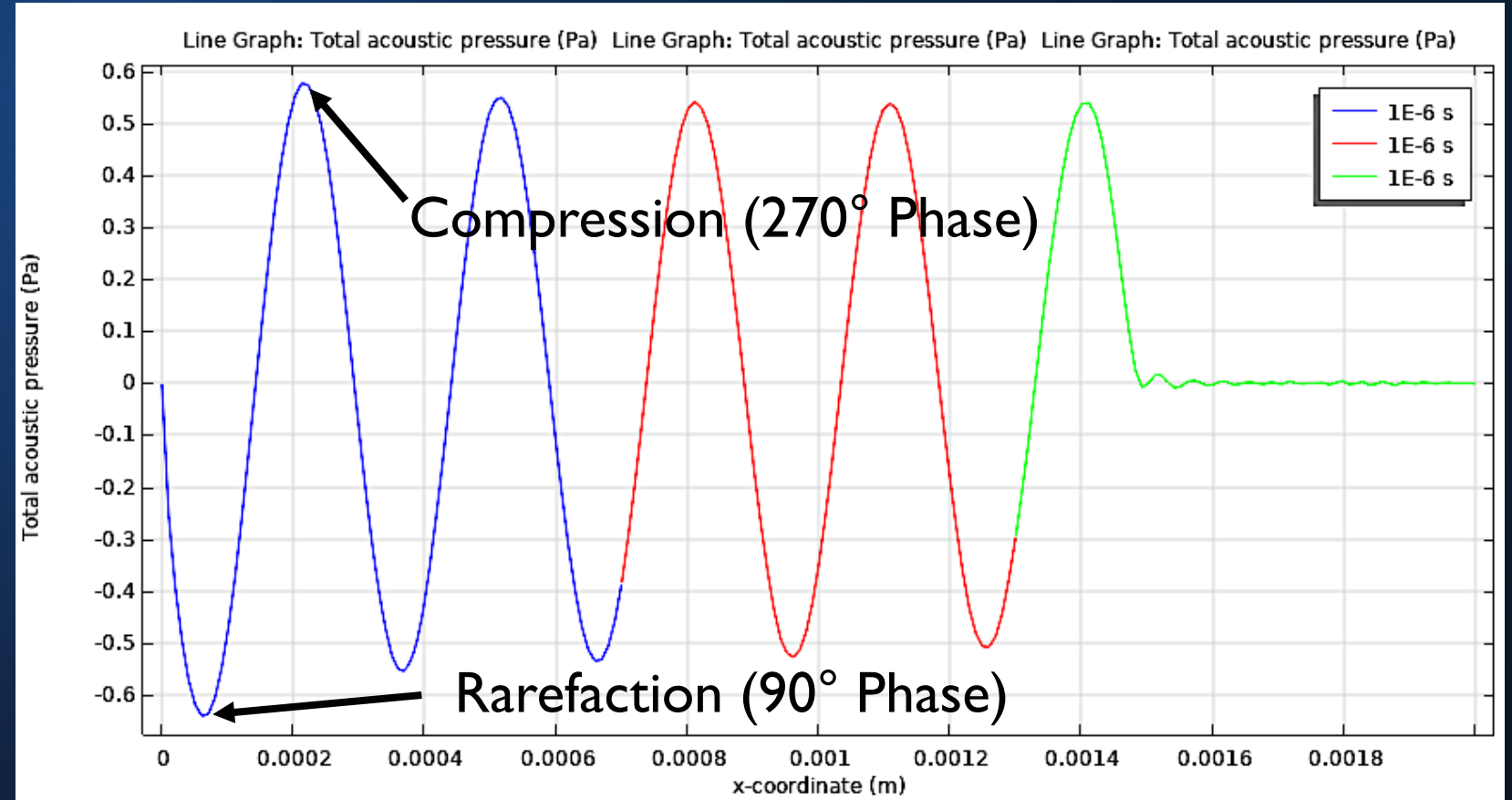


Results – 1 μ s

Total Acoustic Pressure vs. Arc Length



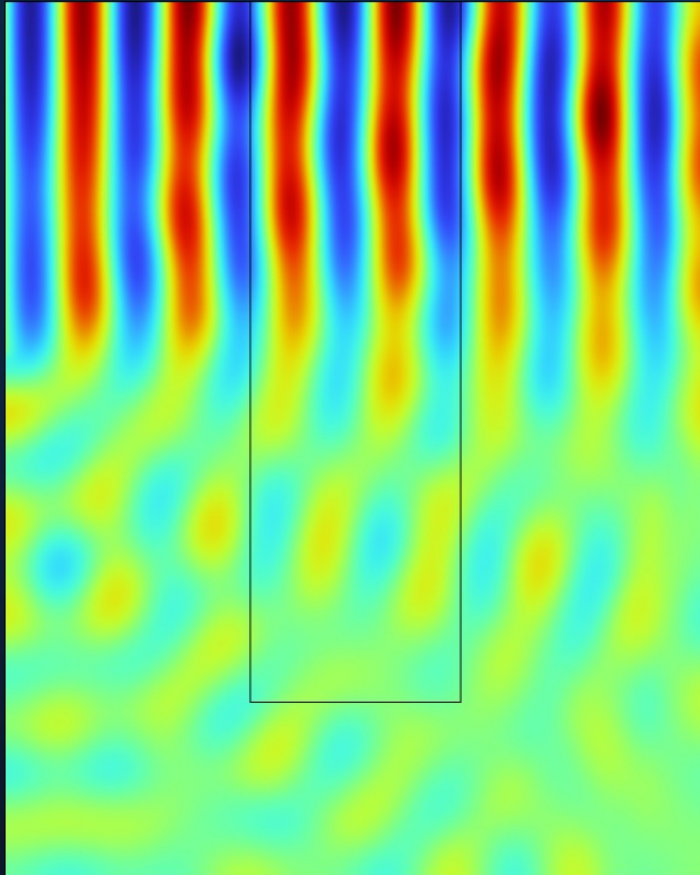
Surface Plot



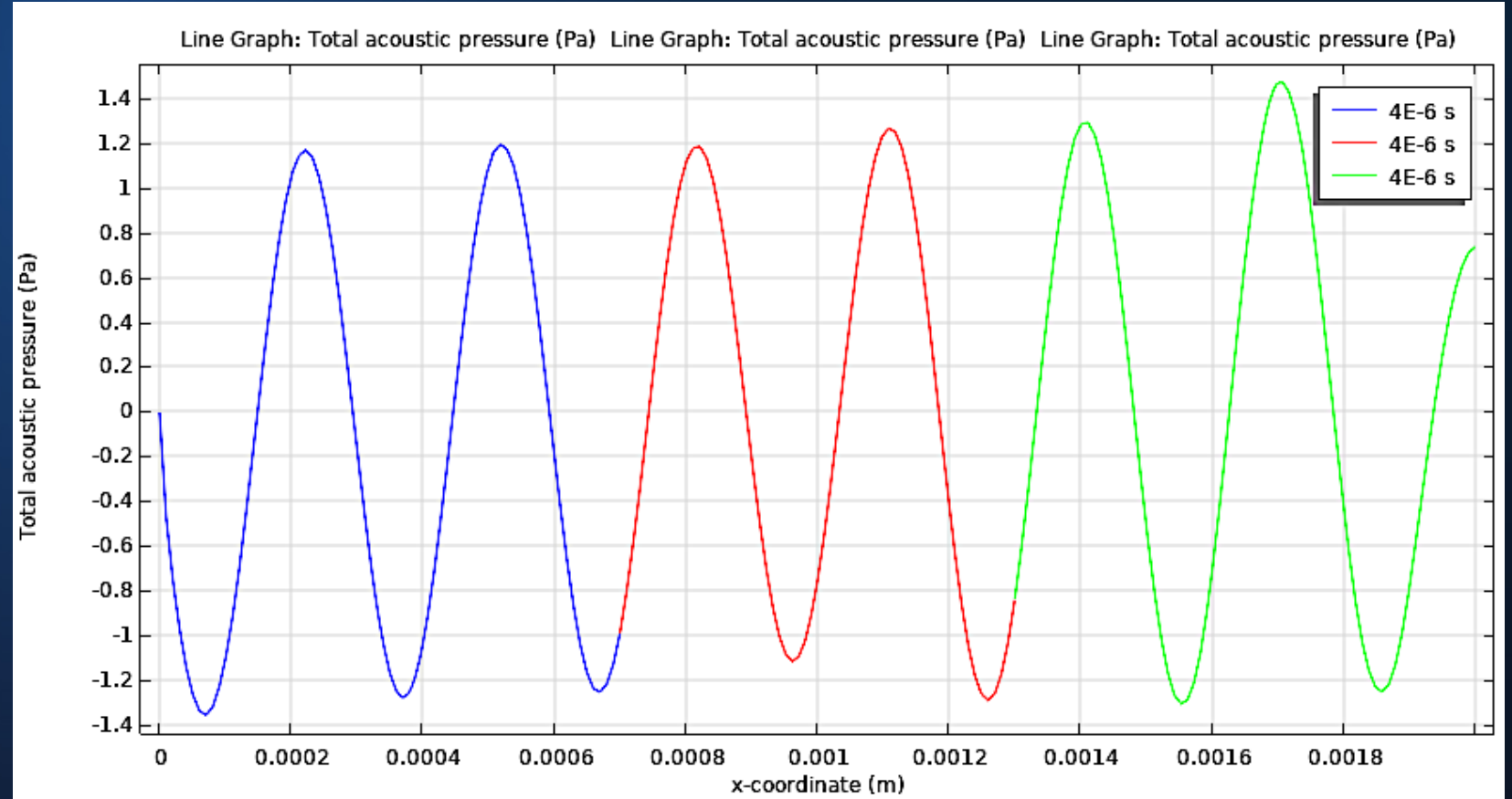
Line Plot

Results – 4 μ s (Control Group)

Total Acoustic Pressure vs. Arc Length



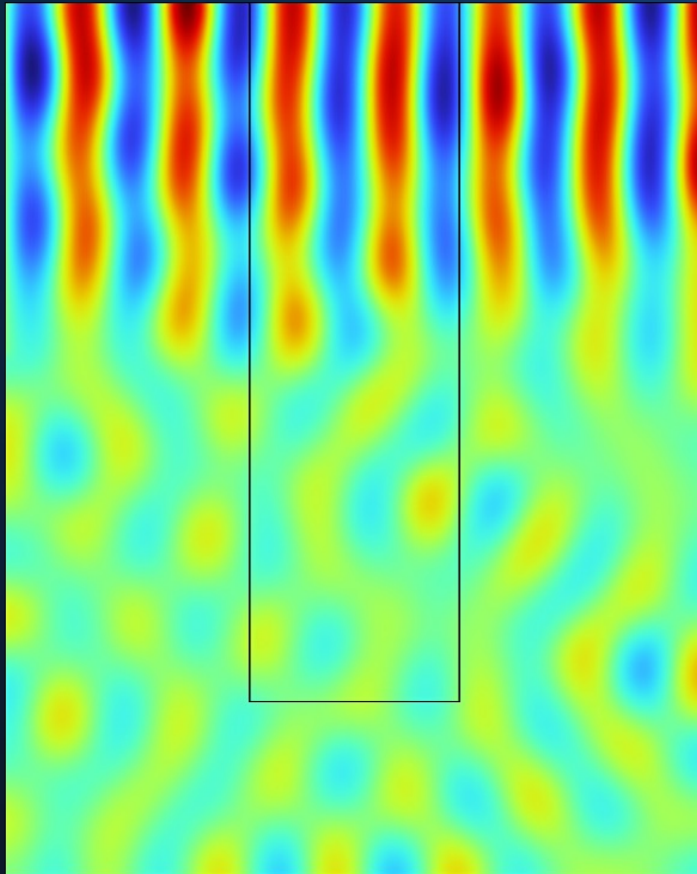
Surface Plot



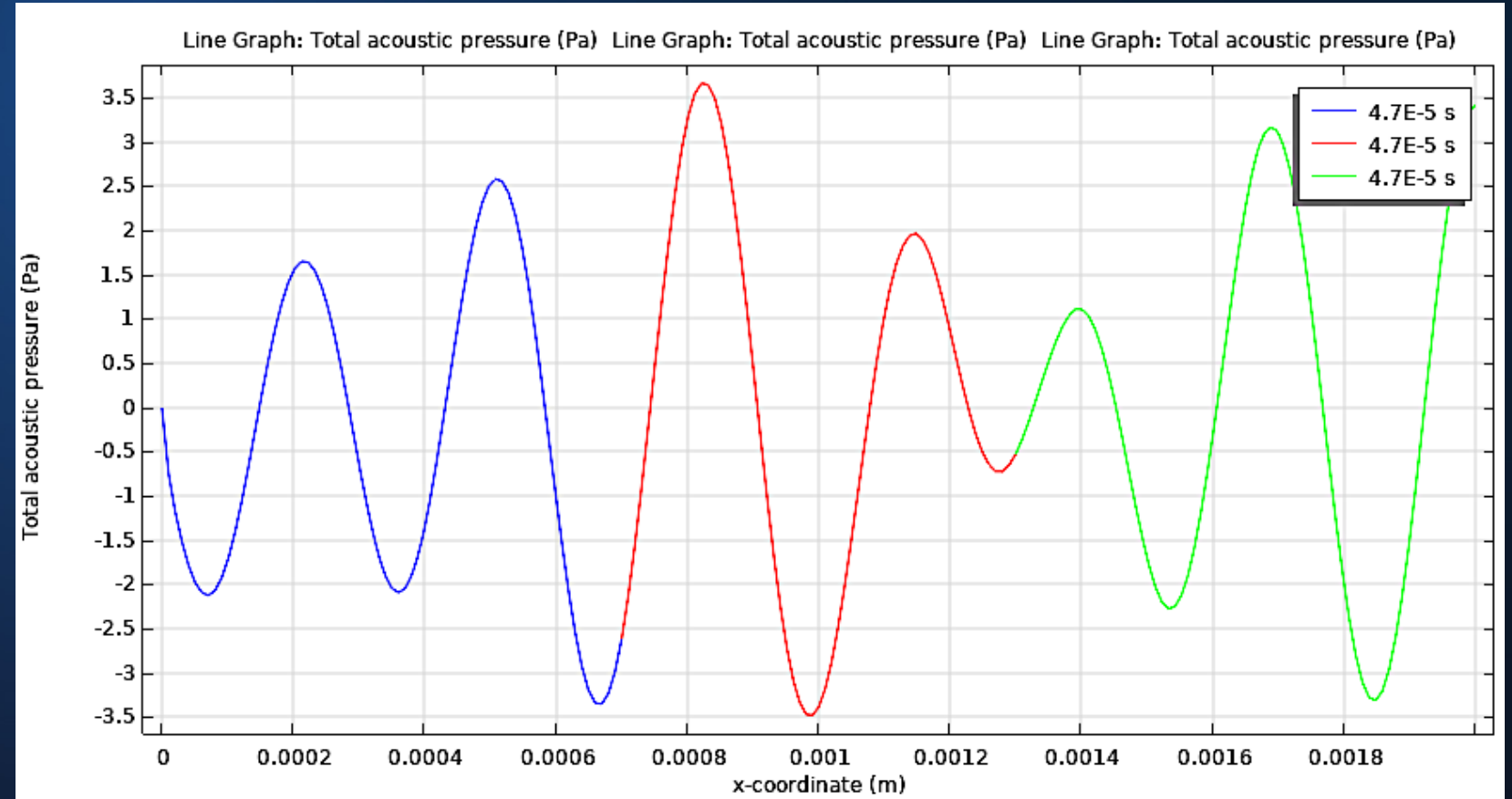
Line Plot

Results – 47 μs

Total Acoustic Pressure vs. Arc Length



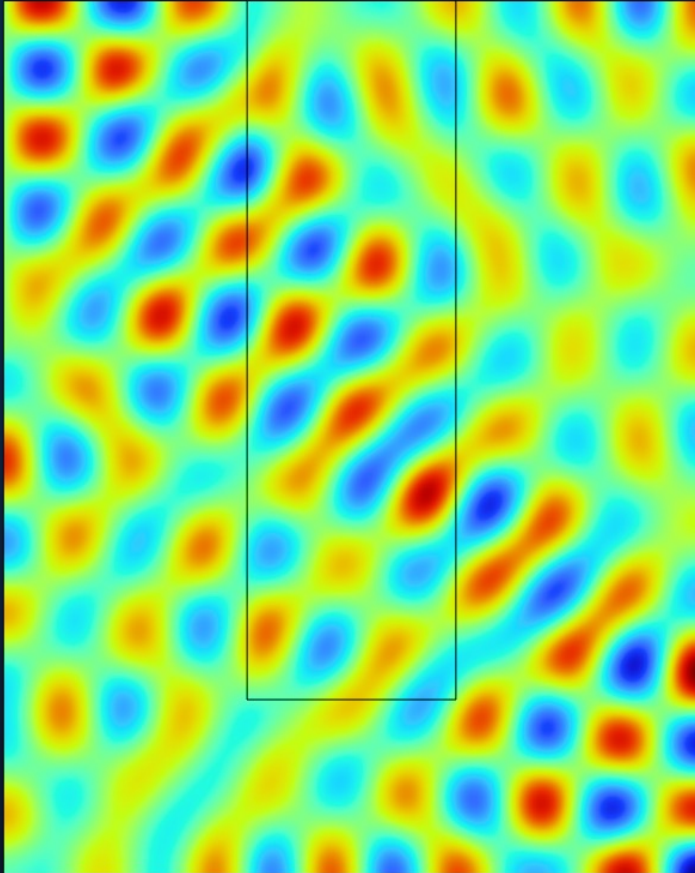
Surface Plot



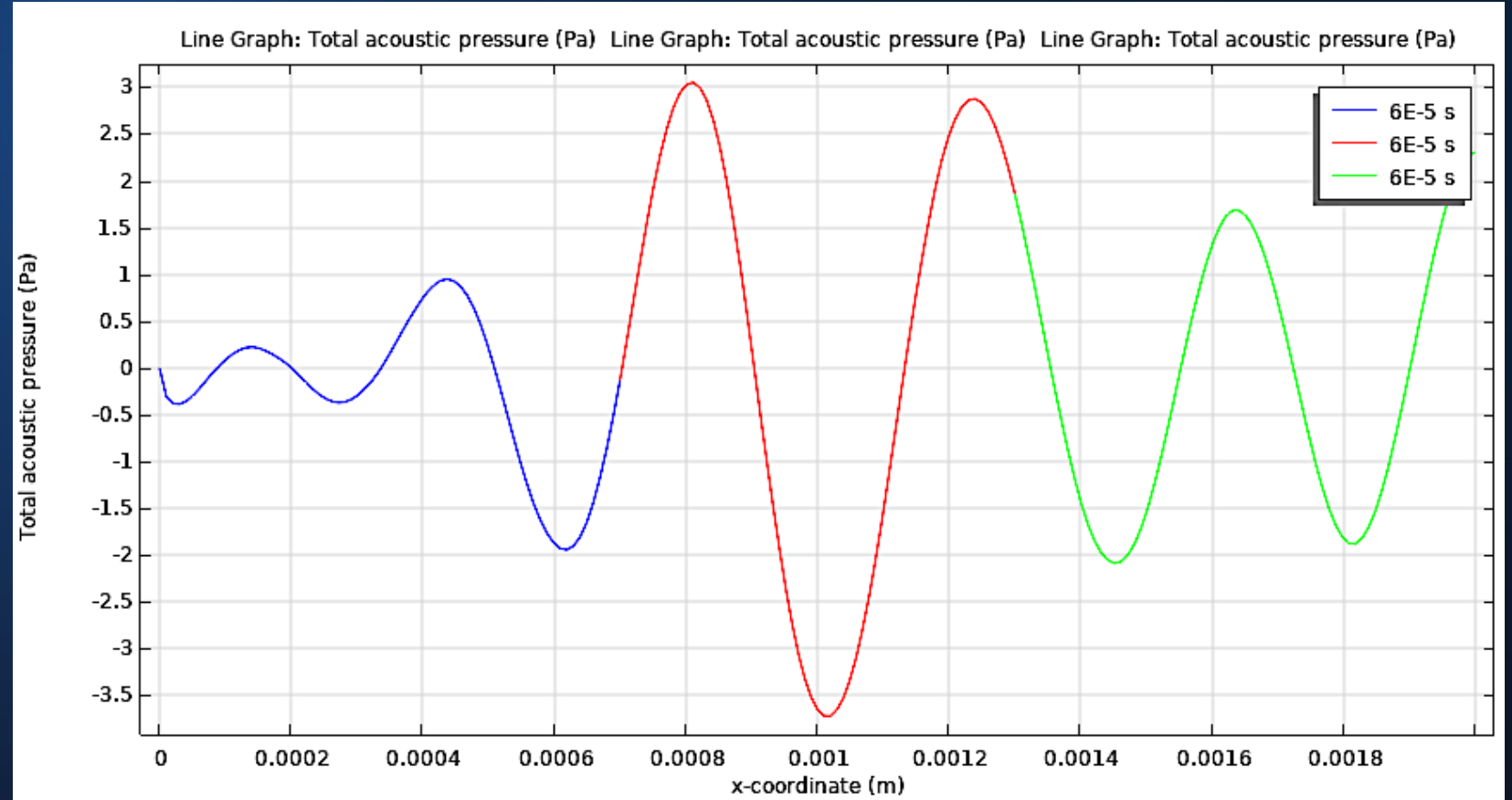
Line Plot

Results – 60 μs

Total Acoustic Pressure vs. Arc Length

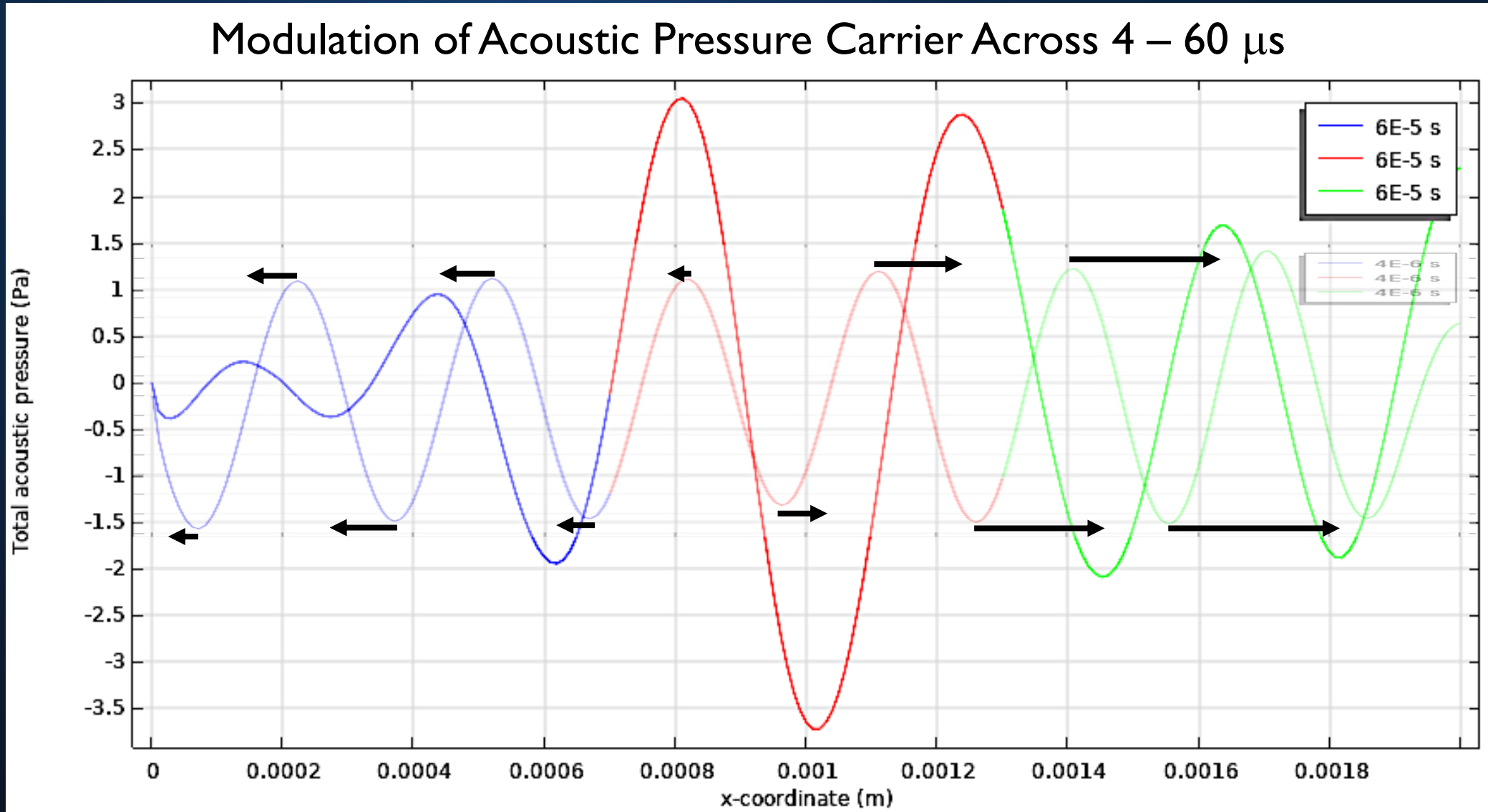


Surface Plot



Line Plot

Results – Total Acoustic Pressure Line Plot



Summary of Results

- Significant Phase Modulation of Ultrasonic Signal
- Slight Amplitude Modulation
 - Close agreement with preliminary results observed in experiment

Conclusions

Problem: Clinical Adoption

- Need to precisely verify dose rate within a volume
- Current dosimetry hardware and measurement standards are lacking

Solution: Novel Detection Principle

- Hardware: use of Ultrasound
- Measurement Standard: Phase Modulation ✓
 - $\Delta\varphi = \Delta(\omega \cdot TOF) \cong \omega \cdot \Delta TOF \cong 18 \text{ mdeg/Gy}$

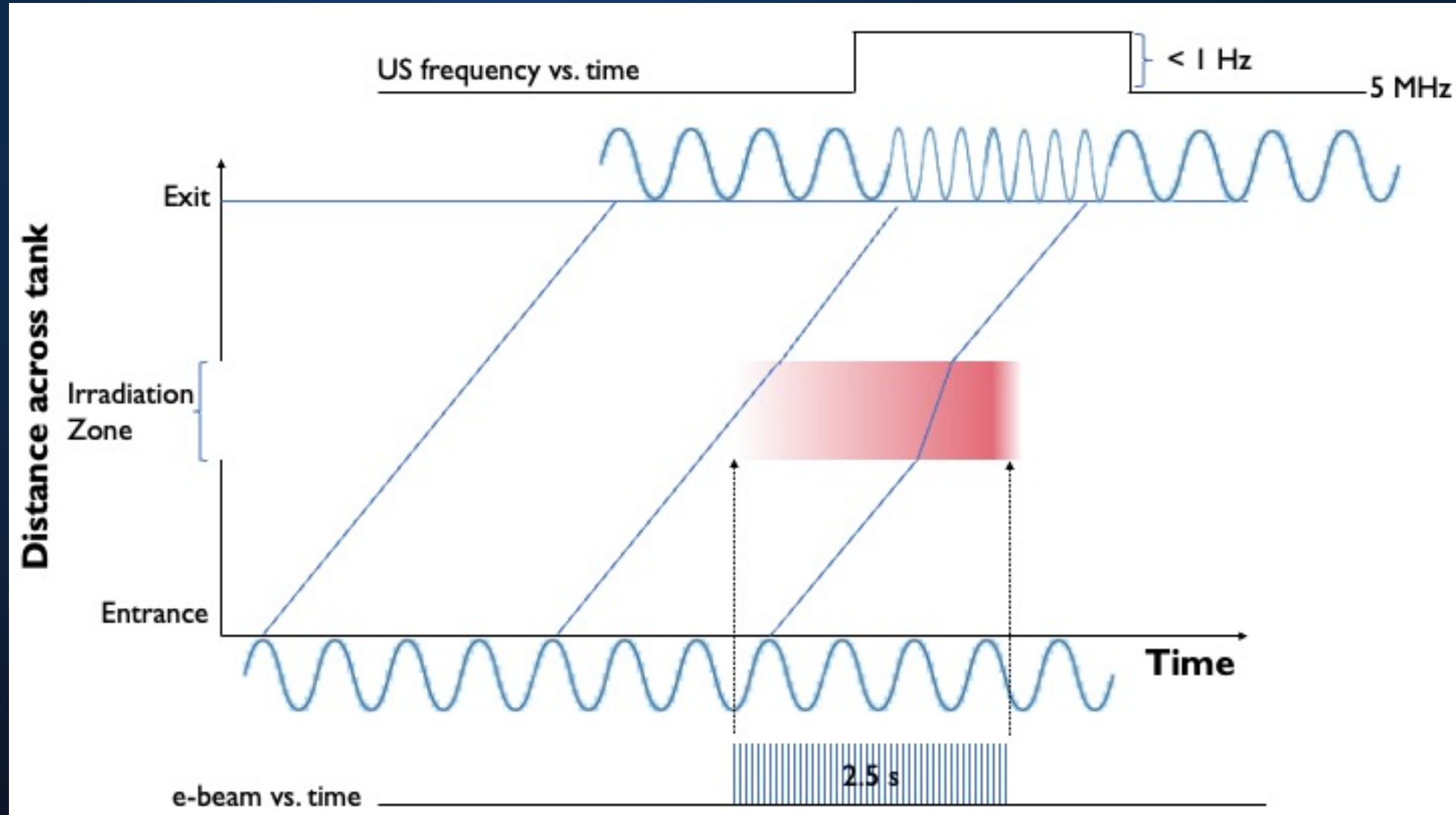
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Appendix A: Experimental Motivations



Appendix B: Experimental Setup

